

Palaeoproteomic characterization of archaeological dental calculus reveals precarious periodontal health in pre-Roman Italy (7th–4th century BCE)

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ABSTRACT

Periodontitis, a chronic inflammatory disease affecting the tooth-supporting structures, is a key indicator of oral health in palaeopathology. While poor oral hygiene, systemic diseases, and genetics are well-established contributors, the dietary impact has often been underestimated. Clinical studies, however, link diets high in fermentable carbohydrates and meat to inflammation. We investigated periodontal disease by analyzing interdental septa in 63 individuals from elite and non-elite groups in pre-Roman Italy (7th–4th centuries BCE), a period of social stratification, intensified agriculture, and increased cereal consumption. Macroscopic analysis was combined with proteomics of dental calculus from 33 individuals. Of the 1890 septa considered, 23 % displayed signs of periodontitis, with significantly higher rates in males. Prevalence increased with age in both sexes. Proteomic findings identified *Porphyromonas gingivalis*, a key periodontal pathogen, in 10 of 19 well-preserved dental calculus samples. While plaque accumulation is the main trigger for periodontitis, our findings highlight the dietary role in disease susceptibility. Carbohydrate-rich foods adhere to teeth and nourish bacteria, worsening periodontal conditions. At the same time, greater access to animal protein, particularly among emerging elites, may have contributed to inflammation. We propose that a proinflammatory diet may have been a major contributor to the proliferation of pathogenic oral microbiota.

1. Introduction

Periodontal diseases are chronic inflammatory conditions that affect the supporting structures of teeth, including the gingiva, periodontal ligament, cementum, and alveolar bone [1–3]. Periodontal disease has been traditionally divided into two categories, gingivitis and periodontitis, based on whether attachment loss occurred [4–7]. The term gingivitis (or dental plaque-induced gingival diseases) is used to designate a reversible stage of periodontal disease since gingivitis is an

inflammatory lesion resulting of general increase in plaque mass around the gingival margin without periodontal ligament and alveolar bone loss [3]. By contrast, periodontitis is a chronic and irreversible inflammatory disease that results from the accumulation of subgingival biofilm (dental plaque), microbial dysbiosis (i.e. imbalanced oral microbiome) with overgrowth of pathogenic bacteria in the gingival crevicular fluid, dental rots and epithelium, and the immune host response leading to destruction of periodontium, which subsequently progresses to alveolar bone loss, resulting in mobility and potentially loss of the affected tooth

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[3,8–13]. Today, it is estimated that 20–50 % of the global population is affected by periodontitis [14] with severe form estimated to affect around 19.0 % of the global adult population, representing the 6th most prevalent condition in the world [15,16].

Molecular studies have shown that the human oral microbiome can consist of up 700 differing taxa, [17] with over 400 of these taxa being identified in the periodontal pockets alone [18]. The traditional 1990's theory of microbial etiology of periodontitis recognized the role of the so-called 'red complex', which comprises Gram-negative anaerobes *Porphyromonas gingivalis*, *Treponema denticola*, and *Tannerella forsythia* as the main bacteria responsible for active periodontitis [19]. However, this model has been reconsidered in 2012 and 2016 under the lens of a synergistic and dysbiotic microbial community rather than by selected 'periopathogens' (i.e., the red complex) and by the concept of *Porphyromonas gingivalis* as the 'key-stone pathogen' responsible of biofilm community of pathogens, respectively. More recently, the IMPEDE model developed in 2020 seeks to explain the nexus between periodontal inflammation and change in the polymicrobial biofilm in the gingival sulcus (for a review [19]).

Dysbiosis in the oral microbiota may be influenced by various non-modifiable and modifiable risks. The former includes host determinants (e.g., immune function, genetics) and menopause, while the latter include smoking, diabetes mellitus, and pregnancy [14,20–22]. Additionally, comorbidities of periodontitis with systemic disorders like cardiovascular disease, rheumatoid arthritis, inflammatory bowel disease are well-established in clinical practice [23–25]; therefore, periodontitis has not only a central role within the realm of oral health, but it represents a major public health issue [26,27].

Among the environment determinants, the role of diet and nutrition towards the expression periodontal diseases is largely overlooked since dietary advice in the dental setting predominantly focuses on the prevention of dental caries [28]. However, balanced macro- and micro-nutrient intake is essential in periodontal health [29]. It has been demonstrated that microbial dysbiosis and periodontal inflammation is not merely due to inappropriate oral hygiene but also includes host environment factors like diets rich in fermentable and/or processed carbohydrates, high-fat dairy products, and saturated fat in meat that promote inflammatory reaction and progression in periodontal diseases [28,30–33], while rich in Omega-3 fatty acids, vitamins, minerals and vegetables may increase periodontal health [28,34,35].

In human archaeological remains, periodontitis is traditionally analyzed by metric measurements with >2 mm threshold of root exposure as indicative of periodontitis [e.g., [36–38,40,41]] against the combination of metric and qualitative alveolar bone evaluation [42–48] or exclusive examination of alveolar bone texture [e.g., [39], [49–54]]. A combined approach of vertical (localized) and horizontal (generalized) bone loss evaluation was applied by Raitapuro-Murray et al. [55]. However, macroscopic evaluation of periodontitis has notable limitations, particularly due to postmortem damage, and individual variation in bone preservation, which can obscure or mimic pathological features and lead to potential misinterpretations. To address these challenges, combining macroscopic analysis with complementary analytical techniques enhances both diagnostic accuracy and the depth of interpretative. In this regard, the application of omic approaches such as metagenomics and proteomics, to dental calculus have only recently provided a new tool to gain insights into periodontitis and to access genetic and protein information of a whole microbial community directly [17,56–60]. Palaeoproteomics of dental calculus is an effective method for investigating both the etiology of and host responses to ancient periodontal disease since subgingival dental calculus formation promote periodontitis [30,61,62]. Proteins are functional agents, and their expression differs in response to pathological conditions. These pieces of evidence, revealing information on oral pathologic processes, could not be obtained solely by DNA analysis, which could only reveal the presence of certain taxa in analyzed specimens. The potential of palaeoproteomics analysis of dental calculus for studying relationship of diet,

health, immune system and oral diseases in the past has not been fully exploited [e.g., [57], [63–65]], despite successful applications in studies aiming at dietary reconstruction in ancient Eurasia [e.g., [66–76]].

In this study, we aim to reconstruct periodontal health in pre-Roman groups from the 1st millennium BCE Italy, a period characterized by social stratification, differential access to resources and intensification of agriculture. This phenomenon was of considerable historical and economic interest [77] and is evidenced by archaeological findings, including the development of complex urban centers, differential burial types and mortuary accoutrements, and definition of rural landscapes shaped by intensive agriculture, with land use focused on staple crops production and livestock husbandry [78–81], as well as by waterworks and discovery of iron tools like ploughshares, sickles, and hoes that points to the technological means for exploiting of arable fields [82]. To explore the biological consequences of these changes, we integrate macroscopic analysis of interdental septa - as indicators of periodontal health - with palaeoproteomic profiling of dental calculus. This combined approach tests the hypothesis that social and economic dynamics of this period are reflected in a proinflammatory diet. Increased reliance on carbohydrate-rich staples as a result of agricultural intensification, as well as greater consumption of animal proteins, often associated with the dietary privileges of emerging elites, may have disrupted oral eubiosis, promoting the proliferation of pathogenic taxa such as *Porphyromonas gingivalis*.

2. Materials and methods

A total of 63 adolescent and adult individuals were available from six elite and non-elite pre-Roman funerary contexts in Italy distributed across three historical regions of Etruria Padana (Bologna Certosa [BC] and Arnoaldi necropolises), Etruria propria, including both northern Etruria (Volterra Le Colombaie [VNC] and Baratti-Fonte San Cerbone [FSC]) and southern Etruria (Tarquinia-Monterozzi-Carraccio Rogani [TM]), and Etruria Campana (Pontecagnano PC)) (Fig. 1) as the last frontier of the Etruscan dominance. All the sites are dated between the 7th and the 4th centuries BCE (Table 1) and detailed descriptions of each site is reported in Supplementary Text 1.

Skeletal sex and age at death estimation were performed on adult individuals included in the present dataset according to well established anthropological methods (Supplementary Text S2). Macroscopic assessment for the presence of periodontal disease and dental calculus deposits was conducted on each individual using appropriate methods (Supplementary Text 3).

2.1. Dental calculus sampling and extraction protein protocol for LC-MS/MS data analysis

Dental calculus was sampled from a subsample of 33 individuals for protein extraction and analysis via liquid chromatography-tandem mass spectrometry (LC-MS/MS). The osteoarchaeological assemblage of individuals of VNC, PC (Table 1) are held at the University of Pisa (Pisa, Italy); those of BC and Arnoaldi necropolises at the Museo Archeologico Civico (Bologna, Italy); those of FSC at the Soprintendenza Archeologia Belle Arti e Paesaggio per le Province di Pisa e Livorno (Pisa, Italy) and those of TM at the Museo Archeologico Nazionale di Tarquinia (Tarquinia, Italy). Permissions to carry out sampling procedures and biomolecular analysis were issued by Museo Civico Archeologico di Bologna (Prot. 2022|117), Soprintendenza Archeologia Belle Arti e Paesaggio per le Province di Pisa e Livorno (Prot 2023|0008350), Parco Archeologico Tarquinia and Cerveteri (Prot. 2022|0000333) and Direzione Regionale Musei Campania (Prot. 2024|0001211), which complied with all relevant regulations.

Between 6 and 15 mg of supra- and/or subgingival dental calculus were removed from each of the 33 individuals. Specifically, calculus was sampled from the mandibular anterior teeth in 18 individuals, from the mandibular posterior teeth in nine individuals, and from the maxillary



Fig. 1. Location of the six archaeological sites across four historical regions in pre-Roman Italy (base map taken from Geoportale Nazionale, Ministero dell’Ambiente, CC BY 3.0).

Table 1
Distribution of six pre-Roman sites sorted by geographical location and corresponding historical region. Numbers of individuals and relevant sample included for macroscopic assessment of periodontal diseases and palaeoproteomics of dental calculus respectively is also reported.

Macro area	Region/province	Archaeological site	Chronology (BCE)	N Individuals	Dental calculus sample for proteomics	Historical Region
Northern Italy	Emilia Romagna/ Bologna	Bologna Certosa (BC)	6th–5th century	8	3	Etruria padana
Northern Italy	Emilia Romagna/ Bologna	Arnoaldi necropolis	7th century	1	1	Etruria padana
Central Italy	Tuscany/Livorno	Baratti - Fonte San Cerbone (FSC)	Second half of the 6th–4th century	6	5	Etruria propria (northern)
Central Italy	Tuscany/Pisa	Volterra - Necropoli Le Colombaie (VNC)	7th–4th century	4	4	Etruria propria (northern)
Central Italy	Lazio/Viterbo	Tarquinia Monterozzi (TM) – Carraccio Rogani	7th -4th century	23	5	Etruria propria (southern)
Southern Italy	Campania/Salerno	Pontecagnano (PC) – Chiancone II, Caramante and Di Renna	7th–4th century	21	15	Etruria Campana
Total Individuals				63	33	

anterior teeth in six individuals, according to well-established recommendations [83] using sterilized dental scalars and stored in sterile 2 ml Eppendorf tubes until extraction. Nitrile gloves were used during sample collection to avoid contamination from skin proteins. Protein extractions were conducted at the Ancient Protein Laboratory at the Institute of Evolutionary Medicine at the University of Zurich (Switzerland).

2.1.1. Protocols

SP3 (single-pot, solid-phase-enhanced) protein extraction to prepare samples for LC-MS/MS analysis was pursued following the protocol published by Wilkin et al. [71]. Following protein extraction, digested peptides were stored at -80°C before being analyzed by liquid chromatography tandem mass spectrometry (LC-MS/MS) at the Functional Genomics Center Zürich (FGCZ) at ETH/University of Zürich.

2.1.2. Protein extraction

Prior to the SP3 protocol, the dental calculus samples are demineralized for 5 to 7 days with 500 μl of EDTA (ethylenediaminetetraacetic acid, 0.5 M) on a rotator at room-temperature (RT). In addition, one extraction blank (Eppendorf tube with 500 μl EDTA) is added to assess the laboratory contamination in the analysis part. For the first day of the SP3 protocol, the samples are removed from the rotator and placed in the centrifuge for 10 min (20'000 rcf, RT). Then, 400 μl of each sample is transferred to clean Eppendorf tubes and stored in the freezer (at -20°C) as a backup. To the remaining samples, 200 μl of GuHCl (guanidine hydrochloride, 6 M) are added and resuspended, which lyses cell walls and denatures proteins into the primary protein structure. After a short centrifugation (15 s, max speed), 30 μl of CAA/TCEP (chloroacetamide/ tris-2(-carboxyethyl)-phosphine) solution (100 mM CAA/ 100 mM TCEP) are added and the Eppendorf tubes are vortexed shortly (~ 2 s) and placed in the thermoshaker for 10 min (0 rpm, 99°C).

After incubation, the samples are removed from the thermoshaker and placed in a rack to cool down for approximately 5 min. Briefly, the samples are centrifuged to get the liquid off the lid (15 s, max speed) and 20 μl (400 μg) of the magnetic hydrophilic and hydrophobic bead solution (20 $\mu\text{g}/\mu\text{l}$) are added by gentle pipetting to each sample. For the SP3-clean up, 350 μl (the same volume of the samples) of 100 % EtOH (ethanol) are added, and the samples are incubated in the thermoshaker for 5 min (1'000 rpm, 24°C).

Then, the Eppendorf tubes are placed in a magnetic rack for up to 2 min, or until the migration of the beads to the magnetic wall has completed. Afterwards, the Eppendorf tubes are removed from the rack, and 200 μl of 80 % EtOH are resuspended with the beads. The tubes are replaced on to the magnetic rack and the total supernatant wash is removed to remove any extraneous non-proteinaceous material. For two additional washes, this step with the 80 % EtOH is repeated, which results in an Eppendorf tube with purified proteins on-beads. For the digestion of the proteins, the Eppendorf tubes are removed from the magnetic rack. Prior to the addition of 100 μl of AmBic (ammonium bicarbonate, 100 mM) solution to each sample, the solution should first be tested for its pH (should be ~ 8). Afterwards, 2 μl of trypsin solution (0.2 $\mu\text{g}/\mu\text{l}$) is added to each sample. The samples are incubated overnight for a maximum of 18 h (750 rpm, 37°C), which avoids an overabundance of trypsin-digested trypsin peptides.

The following day Eppendorf tubes were removed from the thermoshaker and briefly centrifuged (15 s, max speed). Then, the samples are placed in the magnetic rack and the liquid was transferred to clean Eppendorf tubes without transferring the beads. Subsequently, the digestion was halted with the addition of 5 % TFA (trifluoroacetic acid). For lowering the pH of each sample (pH < 2.5), the samples are acidified with a starting volume of 10 μl , and 5 μl are continuously added until the threshold is reached. For the peptide purification using stage-tip clean up, in-house made stage-tips (200 μl) with Empore reversed-phase extraction disks (3 M).

2.1.3. LC-MS/MS setting

Samples were measured on an Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific) equipped with a Nanospray Flex Ion Source (Thermo Fisher Scientific), coupled to an M-Class UPLC (Waters). Solvent composition at the two channels was 0.1 % formic acid for channel A and 0.1 % formic acid, 99.9 % acetonitrile for channel B. Column temperature was at 50°C . For each sample 2 to 5 μl of peptides were loaded on a commercial ACQUITY UPLC M-Class Symmetry C18 Trap Column (100 $\text{\AA}$, 5 μm , $180 \mu\text{m} \times 20 \text{ mm}$, Waters) followed by ACQUITY UPLC M-Class HSS T3 Column (100 $\text{\AA}$, 1.8 μm , $75 \mu\text{m} \times 250 \text{ mm}$, Waters) on a commercial nanoEase. The peptides were eluted at a flow rate of 300 nL/min, after a 3 min initial hold at 5 % B, a gradient from 5 to 22 % B in 80 min and 22 to 32 % B in additional 10 min was applied. The column was cleaned after the run by increasing to 95 % B and holding 95 % B for 10 min prior to re-establishing loading condition. All samples were measured in randomized order.

The mass spectrometer was operated in data-dependent acquisition (DDA) mode with a maximum cycle time of 3 s, using Xcalibur (tune version: 4.4), with spray voltage set to 2.3 kV, funnel RF level at 40 %, heated capillary temperature at 275°C , and Advanced Peak Determination (APD) on. Full-scan MS spectra (350–1200 m/z) were acquired at a resolution of 120,000 at 200 m/z after accumulation to a target value of 3,000,000 or for a maximum injection time of 45 ms. Precursors with an intensity above 5000 were selected for MS/MS. Ions were isolated using a quadrupole mass filter with a 1.2 m/z isolation window and fragmented by higher-energy collisional dissociation (HCD) using a normalized collision energy of 30 %. HCD spectra were acquired at a resolution of 30,000 and maximum injection time was set to Auto. The automatic gain control (AGC) was set to 100,000 ions. Charge state screening was enabled such that singly, unassigned and charge states higher than six were rejected. Precursor masses previously selected for MS/MS measurement were excluded from further selection for 20 s, and the exclusion window was set at 10 ppm. The samples were acquired using internal lock mass calibration on m/z 371.1012 and 445.1200.

2.1.4. Data analysis

Raw MS/MS peptide and protein data files were converted to Mascot Generic Files (MGF) using the msConvert program. Parameters for Mascot were set as follows: digestion enzyme trypsin, carbamidomethyl of cysteine (C) as the fixed modification, variable modifications of deamidation of asparagine (N) and glutamine (Q) and oxidation of methionine (M). Fragment ion mass tolerance for peptide identification was set at 0.01 Da, and peptide mass tolerance was set at 10 ppm. One carbon isotopic shift was allowed for, and up to three missed cleavages. Q-Exactive was selected as the MS/MS instrument. MGFs were searched against a database consisting of Swissprot (downloaded 30.05.2022) and a custom curated database including available, yet unreviewed proteomes from 154 additional plant and animal species found in Eurasia [84] to match the observed spectra to known protein sequences, allowing for additional protein identification possibilities. This database is freely available through Zonodo with the DOI: <https://doi.org/10.5281/zenodo.15805451>. To reduce the number of spurious peptide and protein identifications, resulting peptide identifications were filtered using the R-package MS-MARGE [85], which included only proteins supported by at least two distinct peptide spectral matches (PSMs), and those PSMs with e-values below 0.01. Protein and peptide False Discovery Rates (FDR) were also calculated, with the average peptide FDR of 0.53 %, and average protein FDR of 1.5 %. For the exact FDRs for each sample, please see Supplementary Table S1b. All peptide IDs were also searched with BLAST to confirm their identities and be sure they do not match to other known protein sequences.

Finally, filtered CSV files were compared against the Oral Signature Screening Database (OSSD) set by Bleasdale et al. [86] to establish authentication of ancient proteins, specifically distinguishing between dietary and endogenous oral proteins. This process involves matching the observed spectra to the theoretical spectra derived from the protein

sequences in the OSSD, which includes a subset of oral microbes, human inflammatory response proteins, and common contaminants previously reported in ancient dental calculus samples. The OSSD cannot confirm whether food-related proteins are ancient. Instead, it operates on the premise that the presence of an oral signature suggests that food-related proteins found in archaeological dental calculus are more likely to be genuine ancient rather than contaminants.

All raw data files (.raw and .mgf) and processed files (.mzid) are available in massIVE using the login MSV000097489_reviewer.

2.2. Statistics

Chi-square test was used to provide a test of statistical significance. A *p*-value of <0.05 was used to identify significant differences between factors that in our study are skeletal sex and age, maxilla and mandible. Fisher's exact test was used with the same threshold for statistical significance when expected values were lower than 5. Intra-observer error was quantified using Cohen's kappa and the strength of the agreement was evaluated according to Altman [87]. Data processing and screening was performed with R Studio (R Core Team).

3. Results

The demographic composition of the study osteoarchaeological assemblages is shown in Table 2, with nearly half of the osteoarchaeological assemblage (46.0 %, 29/63) being compromised for allocated age-at-death of the individuals in a discrete age category. Moreover, poor state of preservation prevented assessment of skeletal sex for two individuals (Table 2).

3.1. Macroscopic assessment of periodontal diseases

A total of 810 septa distributed as 447 male septa, 360 female septa, and 3 septa belonging to adults with unknown sex were observed for periodontal diseases. Intra-observer error for Kerr's scoring system was tested for interdental septa of random 35 individuals rechecked at a later date to ensure uniformity of recording. Cohen's κ test showed good agreement between observations according to Altman [87], with κ = 0.710 (CI 0.476–0.954).

Table 3 illustrates the true prevalence of Kerr scores adjusted by the number of unrecordable septa in the males (M), female (F) and individuals with sex unknown (ND). Similar frequencies were observed among the two sexes for both gingivitis (Kerr category 2) while a Chi-square test of adjusted true prevalence rate (TPR) of periodontitis (Kerr categories 3–5) between males (26.0 %) and females (20.9 %) revealed a statistically significant difference (*p* > 0.001) between the two sexes.

In the present study, no septa within a segment were recorded as being entirely healthy (Kerr category 1). The TPR of septa affected by gingivitis (Kerr category 2) and periodontitis (Kerr categories 3–5, Fig. 2) across discrete age groups in the male, female and total samples is reported in Table 4. Septa affected by gingivitis was found to be widespread starting from adolescence onwards with the male subsample significantly more affected than the female counterpart in all the age groups, except for the 35–49 years category. Septa affected by

Table 2
Age and sex composition of the 63 individuals from the six pre-Roman sites.

Age (in years)	Males	Females	ND	Total
13–19	2	2	0	4
20–34	6	5	0	11
35–49	3	7	0	10
50+	6	3	0	9
Adult	15	12	2	29
Total	32 (50.8 %)	29 (46.0 %)	2 (3.2 %)	63

ND = Individuals with sex unknown.

Table 3
Distribution of Kerr categories of male, female and total septa with adjusted TPR. Chi-square outputs deriving from sex comparison with significant values are shown in bold.

Periodontal diseases by sex	Males (%)	Females (%)	ND (%)	Total (%)	Chi-square test
Kerr category 1	38 (4.0)	50 (5.7)	0 (0.0)	88 (4.7)	χ^2 3.1903 df
Kerr category 2	159 (16.6)	128 (14.7)	0 (0.0)	287 (15.2)	1 <i>p</i> = 0.074
Kerr categories 3–5	250 (26.0)	182 (20.9)	3 (5.0)	435 (23.0)	χ^2 11,811 df 1 <i>p</i> = 0.277
Unrecordable septa Kerr category 0	513	510	57	1080	χ^2 6.6397, df 1 <i>p</i> < 0.01
Recordable septa + Unrecordable septa	960	870	60	1890	

ND = septa of individuals with sex unknown.

periodontitis increased across age groups both in the male and female subsamples, with the highest rates in the 50 + years age group. Male septa in the age category 50+ (36.1 %, Table 3) were mostly allocated in the Kerr category 4 (19.4 %), while Kerr category 5 turned to be the highest rate (6.6 %) when compared to the same category across the other age categories (Table 4).

According to inclusion criteria (Supplementary Text 3) for crude prevalence rate (CPR) calculation of periodontitis (Kerr categories 3–5), a total of 45 individuals (25 males and 20 females) were suitable for observation; 72.0 % of males were affected by periodontitis (18/25) and 80.0 % of the females (16/20) with a total of 73.9 % of individuals affected globally.

3.2. Macroscopic evaluation of dental calculus

CPR of dental calculus indicated that 43 out of 55 observable males, females and individuals with sex unknown (see inclusion criteria in Supplementary Text 3) exhibited deposits of dental calculus, with comparable distribution between males (23/29) and females (19/24). As for dental calculus position, more than half of the individuals affected (26/43) had dental calculus exclusively on the mandible, while only five showed calculi limited to maxillary teeth. A combination of deposits on both jaws was observed in seven males and five females. As for severity, the affected male subsample showed a homogeneous distribution of medium (9/23) and slight (11/23) calculus deposits, with only three cases classified as severe. By contrast, females were more prone to develop deposits of calculus in slight degree (11/19) followed by medium degree (7/19), with only case of severe dental calculus deposit observed. Finally, forming position of calculus in the affected 23 males and 19 female individuals revealed equal distribution: supragingival dental calculus was observed in 52.2 vs. 52.6 %, while subgingival dental calculus appeared in 47.8 vs 47.4 %, respectively.

3.3. Palaeoproteomic analysis of dental calculus

Nineteen dental calculus samples from 33 individuals across six sites (57.5 %) passed the Oral Screening Signature Database threshold (Supplementary Table S1a) as described by Wilkin et al. [72].

Bacteria-specific proteins associated to the 'red complex' were identified in 10 subgingival dental calculus samples from six males and four females (Supplementary Table S1b), indicating remarkable presence of periodontitis related-bacteria across four archaeological sites in northern and southern parts of Etruria propria (VNC, FSC, TM sites) as well as in Etruria Campana (PC site). A complete overview of the protein and peptides recovered in dental calculus samples of these 10 individuals is provided in Supplementary Table S2.

Eight individuals out of 10 individuals with identified 'red complex'

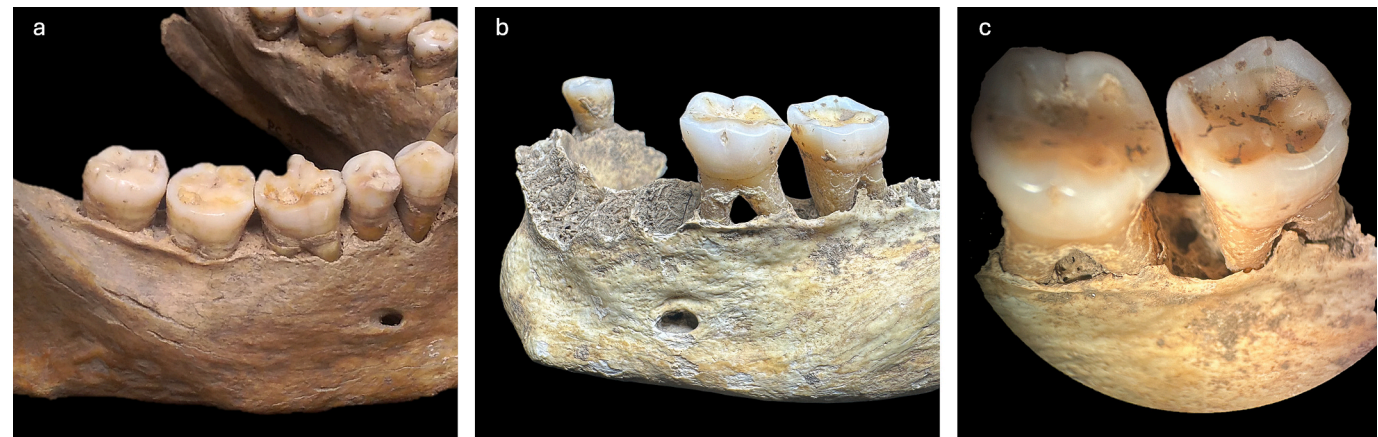


Fig. 2. Palaeopathological examples of periodontitis. a) Kerr category 4: right mandibular molar septa showing loss of normal bone of contour in the form of concave defects with a porous but smooth surface; b) Kerr category 5: deep infra-bony defect between left M₁ and M₂; c) Close-up of the infra-bony defect, showing a honeycombed texture with walls sloping at 45 degrees or more and a depth of at least 3 mm.

Table 4
Adjusted TPR of septa affected by gingivitis (Kerr category 2) and periodontitis (Kerr categories 3–5) in males (M), females (F) and total (Tot) sample allocated in discrete age-at-death categories. Chi-square outputs deriving from sex comparison with significant values are shown in bold.

Periodontal diseases by age	nM (%)	nF (%)	nTot	Chi-square test
Adolescence (13–19 years)				
Kerr category 2	26 (43.3)	15 (25.0)	41 (34.2)	X² 44,829, df 1 p = 0.034
Kerr categories 3–5	5 (8.3)	2 (3.3)	7 (5.8)	Fisher's test p = 0.438
Unrecordable septa – Category 0	9	21	30	
Recordable septa + Unrecordable septa	60	60	120	
Group I (20–34 years)				
Kerr category 2	65 (36.1)	28 (18.7)	93 (28.2)	X² 12.303, df 1, p < 0.01
Kerr categories 3–5	40 (22.2)	34 (22.7)	74 (22.4)	X² 0.0092, df 1, p = 0.92
Unrecordable septa – Category 0	63	75	138	
Observed septa + Unrecordable septa	180	150	330	
Group II (35–49 years)				
Kerr category 2	26 (28.9)	44 (21.0)	70 (23.3)	X² 22.183, df 1, p = 0.136
Kerr categories 3–5	21 (23.3)	64 (30.5)	85 (28.3)	X² 1.583 df 1, p = 0.208
Unobservable septa – Category 0	40	94	134	
Observed septa + Unrecordable septa	90	210	300	
Group III (50+ years)				
Kerr category 2	27 (15.0)	4 (4.4)	31 (11.5)	X² 5.6737 df 1, p = 0.017
Kerr categories 3–5	65 (36.1)	37 (41.1)	102 (37.8)	X² 1.5452 df 1, p = 0.213
Unrecordable septa - Category 0	87	49	136	
Observed septa + Unrecordable septa	180	90	270	

bacteria showed dental septa affected by Kerr categories 4–5 (Supplementary Text 3). We detected evidence of keystone pathogen *Porphyromonas gingivalis* in all the 10 samples. Proteins related to different putative factors of virulence and pathogenicity of *Porphyromonas gingivalis* like gingipains, fimbriae, hemagglutinin, and outer membrane vesicles were identified (Supplementary Table S1b). It is worth noting

that a few sequences (Supplementary Table S1b) are shared between both human periodontal pathogen *Porphyromonas gingivalis* and its corresponding Canidae type called *Porphyromonas gulae*. Of the samples containing these bacteria, the protein false discovery rate (FDR) ranged between <0.01 to 2.41, with an average of 1.5, and peptide FDRs ranged from <0.01 to 1.44, with an average of 0.528.

4. Discussion and conclusion

4.1. The role of sex-steroid hormones and host immune function in periodontal diseases

The CPR of gingivitis (Kerr category 2) in our study was not statistically difference between the sexes, with male and female individuals similarly affected. By contrast, when considering the TPR of periodontitis, male septa were statistically more affected than the female counterpart. Modern epidemiological studies have identified higher prevalence of periodontitis in men compared to women with sex differences explained considering a combination of behavioral, social and biological factors responsible for higher susceptibility to periodontitis in males [88,89]. Males tend to have poorer oral health practices compared to females, and detrimental oral health status in men with low socio-economic status [88].

Biologically, it is well established that male immune systems are at a disadvantage compared to females, who exhibit stronger cellular and humoral immune responses [90–93]. Epidemiological studies have shown that males are more prone to severe periodontal disease than females, partly because of androgens on immune response and tissue metabolism [88,89,94]. Emerging clinical research indicates that androgen hormones in males could influence periodontal health and disease. Testosterone has been shown to influence the function of neutrophils and macrophages, key players in the body's defense against periodontal pathogens. High levels of testosterone have been associated with altered levels of cytokines responsible for more aggressive inflammatory immune response compared to females [95]. While testosterone has been found to have a protective role in maintenance of cancellous bone mass [96], current human and animal studies on the relationship between testosterone and periodontitis reveal discordant findings with acceleration of alveolar bone loss with both excessive high and low level of testosterone [97–101].

Periodontitis is a chronic disease, the effects of which accumulate with age. It is therefore reasonable to assume that untreated gingivitis can progress to periodontitis through life and would have a greater impact on older patients compared with younger patients. In this context, the pre-Roman adults over 50+ years who exhibited the highest

rates of periodontitis in both sexes and gingivitis of the male septa could be interpreted considering the concept of immunosenescence. Immunosenescence, i.e. the age-related decline in immune function plays a crucial role in the increased susceptibility and severity of periodontitis in older adults [102–104]. The reduced functionality of neutrophils and macrophages results in less effective clearance of periodontal pathogens. The chronic systemic inflammatory state, termed inflammaging, exacerbates the inflammatory response in the periodontal tissues, leading to greater tissue destruction and bone loss [104,105]. Immunosenescence may alter the composition of the oral microbiome, favoring the growth of pathogenic species associated with periodontitis. This shift in microbial communities can further drive the inflammatory response and tissue breakdown [104]. However, periodontal diseases do not strictly debut during adulthood, but also during childhood and adolescence [11]. In our study, gingivitis was found widespread starting from the 13–19 age category onwards (Table 4). According to clinical literature periodontal diseases are among the most frequent diseases affecting children and adolescents. These include gingivitis, localized or generalized prepubertal periodontitis and juvenile periodontitis and periodontal diseases associated with systemic disorders [106]. Plaque-induced chronic gingivitis is the most prevalent form due to improper oral hygiene practices among juveniles while steroid hormone related gingivitis is a common periodontal infection of adolescents due to high levels of sex steroid during puberty [106,107].

4.2. Proinflammatory diet as modifiable factor responsible for periodontal disease

Populations lacking regular dentifrice techniques or access to systematized oral healthcare professionals often exhibit mild to extreme amounts of supragingival and subgingival calculus, strongly associated with periodontal diseases [108]. In the present study, deposits of both supragingival and subgingival calculus, resulting from the mineralization of supra- and subgingival dental plaque, were found in nearly all the pre-Roman individuals observed (see Results). This was not unexpected given the rudimentary or neglected oral hygiene practices in ancient societies. The subgingival type in particular facilitates the accumulation and retention of subgingival dental plaque which is a primary contributor to periodontal disease, as they release toxins and enzymes that cause inflammation and tissue destruction [11]. The rough surface of subgingival calculus supports the formation and maturation of a complex biofilm that harbors periodontopathogens such as *Porphyromonas gingivalis* which is crucially involved in chronic periodontitis [109]. The presence of subgingival calculus and the associated bacterial biofilm elicits a chronic inflammatory response from the host [8,110,111].

However, poor oral hygiene and lack of dental treatments are not the only extrinsic and modifiable factors responsible for dental plaque formation and its dysbiosis [112]. Diet has been shown to influence the microbial composition and diversity of saliva and gingival crevicular fluid [113]. Plaque-encouraging nature of the diet, such as that based on excessive intake of fermentable carbohydrates as well as farmed animal meats are recognized as a crucial risk factor for oral biofilm dysbiosis and the increase of both dental caries and periodontal disease development [114–116], with keystone pathogen of *Porphyromonas gingivalis* as indicator of periodontal dysbiosis [19]. With this regard, excessive intake of protein-rich foods, whether plant-based or terrestrial animal sources, can also negatively affect the host's immune response. High blood glucose levels, often associated with excessive carbohydrate-rich diet, along with elevated consumption of meat high in saturated fats, can both induce oxidative stress and the accumulation of advanced glycation end products, resulting in a hyper-inflammatory state that contributes to the destruction of periodontal tissues [e.g., [115–117]].

Our study revealed oral microbiota dysbiosis, enriched in periodontitis-associated bacteria, in 33.3 % of the subgingival dental calculus samples analyzed ($N = 33$). Although dental calculus samples in the present study related to human species only, we cannot totally rule

out a possible transmission of periodontal pathogens deriving from intense human-dogs interaction in five individuals (Supplementary Table 1b), a risk that has been detected in recent clinical literature [118].

Affected individuals exhibited dental septa affected by periodontitis and categorized as Kerr category 3 in two cases, Kerr category 4 in five cases, and Kerr category 5 in five cases. Due to their multifactorial nature, dental caries and periodontitis are not mutually exclusive. It is not surprising that a susceptible host with a weak immune response and infrequent exposure to fluoride may suffer from both periodontitis and dental caries in the presence of a carbohydrate-rich diet. Dental caries affected 9.4 % of teeth in the pre-Roman male subsample ($n = 544$) and 13.5 % in the female subsample ($n = 430$), with an overall prevalence of 11.1 % ($N = 997$, including individuals of unknown sex). Cooccurrence of dental cavities was observed in 13 of the 18 male individuals affected by periodontitis (Kerr categories 3–5) and in 12 of the 16 female individuals affected by periodontitis (Kerr categories 3–5) [119].

Dental caries and periodontitis are the most common oral diseases, and although they are often considered separately, there are correlations between the two conditions [115], with recent studies have demonstrated how a carbohydrate-rich diet promotes microbiota disruption in the human oral cavity, inducing an inflammatory reaction and causing periodontitis [32]. Dental caries is caused by demineralization of dental tissue by acids produced by cariogenic bacteria, such as *Streptococcus mutans* and *Lactobacillus* during carbohydrate fermentation [120], changing the composition of the oral microbiome [121]. The dysbiosis as a result of dental caries creates an environment favorable to the proliferation of anaerobic pathogenic bacteria such as those of the red complex [122]. These bacteria thrive in the anaerobic conditions such as the subgingival region and dental pockets where they are able to survive in areas of complex nutritional provision and reduced oxygen; here require iron in the form of heme; an inability to ferment carbohydrates; and an obligately anaerobic respiration [123–125] and responsible for perpetuating a cycle of inflammation and tissue destruction that characterizes periodontitis. The idea that bulk of the diet relied on increased consumption of soft carbohydrate-rich foods is supported by previous bone collagen stable isotope analyses of certain Etruscan groups from the site of Pontecagnano [126], which are also part of the current study (see Supplementary Text 1). These analyses revealed that the adult diet in these groups predominantly consisted of C₃ plants, with limited intake of terrestrial protein; by contrast, elite members from sites in Etruria Padana (BC and Arnoaldi, see Supplementary text 1) and some of the individuals from Etruria propria (FSC, TM see Supplementary Text 1) exhibited isotope profiles consistent with pro-inflammatory diet marked by a high intake of terrestrial animal proteins [119].

In palaeopathology, periodontitis is generally discussed solely based on oral hygiene and only sporadically in relation to age and skeletal sex or considering the intersection with diet [e.g., [39]]. By incorporating factors such as diet into the discussion, a more comprehensive understanding can be achieved. For instance, Leigh [127] showed differences between groups of Native Americans with varying patterns of subsistence, finding that bone loss was most marked in those with the greatest reliance on maize agriculture. Similarly, Fyfe et al. [128] found high levels of alveolar bone loss among 16th–17th century CE Solomon Islanders, which they interpreted in relation to carbohydrate consumption. More recently, Whiting et al. [51] considered the increase in crop consumption and its potential impact on oral flora diversity to contextualize the varying rates of periodontal disease in Neolithic to Medieval osteological assemblages from Sudan. Additionally, Pederngana and Huber [54] hypothesized that the precarious periodontal status in some of the analyzed individuals from early medieval Switzerland could be attributed to vitamin deficiencies.

The combined approach of omic technologies to archaeological dental calculus has begun to reveal the oral microbial communities of past human groups offering a unique opportunity to advance our knowledge of the bacterial and periodontal status of the oral cavity and

how dietary changes directly impacted the human oral microbiome [e.g., [57,64,129,130]]. In this regard, oral dysbiosis was identified during large scale periods of significant dietary shifts in human evolution, including the adoption of a carbohydrate-rich diet during the Neolithic [63,131], the Bronze Age [132], and the Industrial Revolution in the 18th century CE [e.g., [59,63,133]]. During these times, the oral microbiome shifted towards cariogenic bacteria, and farming populations exhibited more periodontal disease-associated taxa, including *Porphyromonas gingivalis*.

The addition of proteomics analysis of dental calculus has significantly advanced our understanding of oral health in pre-Roman Italy. The challenges of studying periodontal disease through a macroscopic approach lie in the methodological limitations of such studies, which may overestimate the prevalence of periodontitis in skeletal assemblages due to the reliance on purely metric evaluations [134]. Even when qualitative assessments of septa are pursued, as in this study, they are not free from caveats. Post-mortem damage can have differently impacted the preservation of alveolar bone of the maxilla and mandible, leading to incomplete or distorted evidence of periodontal disease. Similarly, different individuals or teeth within the same individual may have experienced varying levels of preservation, potentially skewing the interpretation of disease prevalence and severity. The integration with proteomics has allowed us to analyze preserved proteins within archaeological dental calculus, providing a direct and unbiased look into the oral microbiome and host-pathogen interactions during a key period of pre-Roman Italy. This molecular approach has offered insights into how ancient individuals' bodies responded to oral infections, giving a more nuanced understanding of the chronicity and severity of oral diseases.

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CRedit authorship contribution statement

Giulia Riccomi: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Lisa Rosselli:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Marinella Marchesi:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Federica Guidi:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Maria Taloni:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Giovanni Ricci:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Carmine Pellegrino:** Writing – review & editing, Writing – original draft, Resources, Formal analysis. **Shevan Wilkin:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Data availability

All relevant biomolecular data are within the paper and its Supporting files. Accession number to massIVE: MSV000097489.

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